

Four problems — answers

Item 1 is worked in full on the handout.

2 — The same gate at $n = 3$

$$x_{IH} = 5.80, \quad y_{OH} = y(0.80) = 6.65, \quad y_{OL} = y(5.80) = 0.15.$$

The band narrowed, and that is good. At $n = 2$ the uncertain region runs from 1.02 to 9.80, a factor of 9.6. At $n = 3$ it runs from 0.80 to 5.80, a factor of 7.3. The output levels also moved apart, 6.65 against 0.15 rather than 4.95 against 0.20.

Both changes help the same thing: a sharper gate has a narrower band of inputs it cannot classify, and it pushes its outputs further from that band. That is why the next item is easier to solve for a sharper gate, and it is the first half of the session's trade — the second half is item 4.

3 — Does this pair connect?

A's output levels are $y_{OL} = 0.20$ and $y_{OH} = 4.95$ (item 1). B's thresholds are $x_{IL} = 0.80$ and $x_{IH} = 5.80$ (item 2).

$$NM_L = \log_{10} \frac{0.80}{0.202} = +0.60, \quad NM_H = \log_{10} \frac{4.95}{5.80} = -0.07.$$

The high side fails, by 0.07 decades — about 17%. A's best HIGH output is 4.95 and B does not read anything below 5.80 as HIGH. The pair is a hair short, which is the common case in practice and the reason the test is worth doing with numbers rather than by eye: nothing about these two curves looks wrong.

The RBS window. Scaling A's output by f multiplies both of its levels by f and moves neither gate's thresholds:

$$NM_H \geq 0 \iff f \geq \frac{5.80}{4.95} = 1.17, \quad NM_L \geq 0 \iff f \leq \frac{0.80}{0.202} = 3.96.$$

So any factor between **1.17 and 3.96** works — a 3.4-fold window — and the most tolerant choice is its geometric middle, $f = 2.15$, which gives both margins **+0.26** decades.

Worth noticing against item 2: B is the *sharper* gate, and sharpening the receiver is what made this pair nearly work. The identical-gate pair on the slides, $n = 2$ driving $n = 2$, fails by 0.30 decades; replacing the receiver with the $n = 3$ gate cuts that to 0.07 before any RBS is touched.

Why the RBS does not move A's thresholds. Multiplying $y_{\min}$ and $y_{\max}$ by the same factor leaves $(y - y_{\min})/y$ unchanged at every x , so $|G|$ is unchanged, so the two solutions of $|G| = 1$ are unchanged. The RBS moves what a gate emits; the promoter's K and n set what it reads.

One caveat, and it is about real gates rather than this model. Here the output is a reporter with its own ribosome binding site, separate from the repressor's. In a repressor chain the output protein *is* the next gate's repressor, so a single RBS sets both what the gate emits and the level at which the next gate reads it. Nielsen et al.'s supplement (section I.C) report exactly that: changing the RBS controlling repressor expression shifted the gate's threshold, and in some cases eliminated the response. Two knobs here; often one knob in the laboratory.

4 — A sensor that must read four decades

Why one promoter cannot. A Hill curve is a straight line on a log axis only near its midpoint. The same slope that gave us gain gives us this: on a log axis the slope of $u^n/(1 + u^n)$ is $ny(1 - y)$, largest at $y = 1/2$. Staying within 75% of that peak keeps y between $1/4$ and $3/4$, and the input ratio between those two points is

$$9^{1/n} : \quad 9 \times \text{ at } n = 1, \quad 3 \times \text{ at } n = 2, \quad 1.7 \times \text{ at } n = 4.$$

Four decades is ten thousand-fold. Even the gentlest useful promoter, $n = 1$, gives you not quite one decade. And the cooperativity that made item 2's gate a better switch makes it a worse sensor — the same n , in both bills, with opposite signs.

What to change. Both ends of the curve saturate, so remove both saturations. Daniel et al. (*Nature* 2013) do exactly that: positive feedback on the transcription factor, so that rising inducer makes more factor and never saturates it, plus a high-copy plasmid of binding sites — a shunt — that soaks up free factor so the DNA sites never saturate either. Their measured output fits $\ln(1 + x)$ over more than three decades (Fig. 1d), where the open-loop control on the same axes is fitted by $x/(1 + x)$ and spans a narrow range.

Other answers that earn full credit: several promoters of different K reading the same inducer, with their outputs combined (a segmented sensor, which is how most instruments do it); or a slow integrator that converts concentration into a time, trading range for speed. Any answer that names which saturation it removes is doing the right kind of reasoning.